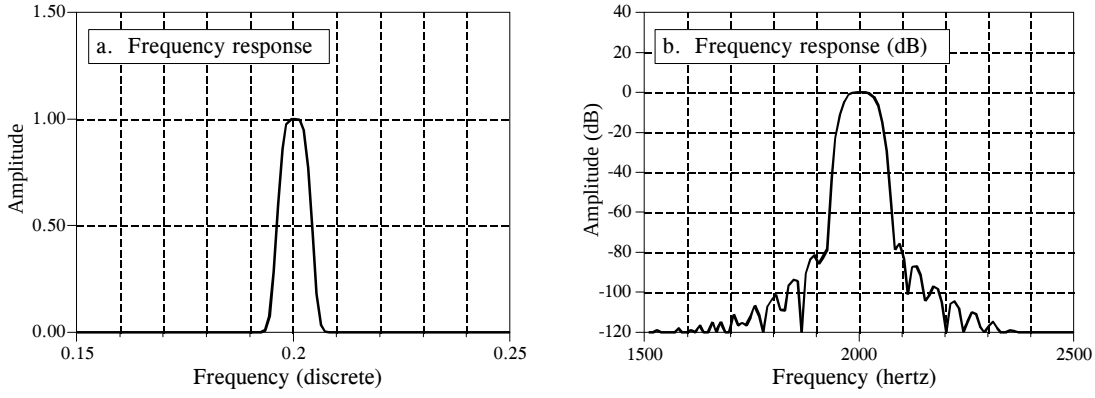

FIGURE 16-6
Example of a windowed-sinc band-pass filter. This filter was designed for a sampling rate of 10 kHz. When referenced to the analog signal, the center frequency of the passband is at 2 kHz, the passband is 80 hertz, and the transition bands are 50 hertz. The windowed-sinc uses 801 points in the filter kernel to achieve this roll-off, and a Blackman window for good stopband attenuation. Figure (a) shows the resulting frequency response on a linear scale, while (b) shows it in decibels. The frequency axis in (a) is expressed as a fraction of the sampling frequency, while (b) is expressed in terms of the analog signal before digitization.

of the sampling rate. The transition bandwidth will be set at 4 hertz, or 0.04 of the sampling rate. From Eq. 16-3, the filter kernel needs to be about 101 points long, and we will arbitrarily choose to use a Hamming window. The program in Table 16-1 shows how the filter is carried out. The frequency response of the filter, obtained by taking the Fourier Transform of the filter kernel, is shown in Fig. 16-5.

In a second example, we will design a *band-pass filter* to isolate a *signaling tone* in an audio signal, such as when a button on a telephone is pressed. We will assume that the signal has been digitized at 10 kHz, and the goal is to isolate an 80 hertz band of frequencies centered on 2 kHz. In terms of the sampling rate, we want to block all frequencies below 0.196 and above 0.204 (corresponding to 1960 hertz and 2040 hertz, respectively). To achieve a transition bandwidth of 50 hertz (0.005 of the sampling rate), we will make the filter kernel 801 points long, and use a Blackman window. Table 16-2 contains a program for calculating the filter kernel, while Fig. 16-6 shows the frequency response. The design involves several steps. First, *two* low-pass filters are designed, one with a cutoff at 0.196, and the other with a cutoff at 0.204. This second filter is then *spectrally inverted*, making it a high-pass filter (see Chapter 14, Fig. 14-6). Next, the two filter kernels are added, resulting in a band-reject filter (see Fig. 14-8). Finally, another *spectral inversion* makes this into the desired band-pass filter.

Pushing it to the Limit

The windowed-sinc filter can be pushed to incredible performance levels without nasty surprises. For instance, suppose you need to isolate a 1 *millivolt* signal riding on a 120 *volt* power line. The low-pass filter will need